Project Title

News and Social Media Analytics Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
N Charishma (2520090214.)
Kaveri (2520090068.)

Team Number: 30

Section:12

Under the guidance of

<<K Chandusha>>

<<Professor, CS&IT>>



Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "***News and Social Media Analytics Engine***" is a record of Bonafide work of **N Charishma – 2520090214., Kaveri – 2520090068.** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

N Charishma – 2520090214.

Kaveri – 2520090068.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled "***News and Social Media Analytics Engine***" is a bonafide work done and submitted by **N Charishma – 2520090214., Kaveri – 2520090068.** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, <<Guide Name>>, <<Designation>>, <Department>> which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

N Charishma – 2520090214.

Kaveri – 2520090068.

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	6
2.	Introduction	7
3.	Literature survey	8
4.	Problem Statement and Objectives	11
5.	Hardware and Software requirements	12
6.	Implementation	13
7.	Results & Discussion	18
8.	Conclusion & Future Scope	19
9.	References	20

1.ABSTRACT

The **News & Social Media Analytics** project aims to analyze news and social media content to understand public sentiment and identify trending topics. As large amounts of textual information are generated daily, manually examining and interpreting this data can be time-consuming. This project provides a simple web-based application that organizes and analyzes textual data from a CSV dataset.

The system is developed using **Java for backend processing and HTML, CSS, and JavaScript for the frontend**. The dataset contains information such as news content, source, category, sentiment, and score. The system loads the dataset, preprocesses the text, and applies a rule-based approach to classify content into three sentiment categories: Positive, Negative, and Neutral. It also identifies frequently occurring keywords to determine trending topics within the available dataset. The processed information is provided through Java-based APIs and displayed on an interactive web dashboard.

The application includes a news feed, sentiment analysis, trending topics, and an analytics dashboard. These features help users understand sentiment distribution, explore individual records, and identify recurring topics in the collected data. The project demonstrates how basic text processing and data analysis techniques can be integrated into a web application to simplify information interpretation. Overall, the system provides a user-friendly platform for analyzing news and social media data. In the future, it can be enhanced by integrating live data sources, machine-learning models, and advanced visualization techniques to improve analytical capabilities.

Team Members:

1. N Charishma, 2520090214
2. Kaveri, 2520090068

2. INTRODUCTION

In today's digital world, news websites and social media platforms generate large amounts of textual information every day. This information contains public opinions, reactions, and discussions about various topics, events, and issues. Analyzing such data can help users understand the general sentiment of the content and identify topics that frequently appear. However, manually reviewing large amounts of text is time-consuming and makes it difficult to identify patterns efficiently.

The **News & Social Media Analytics** project is developed to simplify the analysis of news and social media content using text processing and data analysis techniques. The system uses a CSV dataset containing news records and related information, such as source, title, content, category, sentiment, and score. The data is processed using Java, while HTML, CSS, and JavaScript are used to develop a user-friendly web interface.

The main purpose of this project is to classify textual content into Positive, Negative, and Neutral sentiment categories and identify trending topics based on frequently occurring keywords in the dataset. The application presents the analyzed information through a web dashboard that includes a news feed, sentiment analysis, trending topics, and analytical summaries. This enables users to explore the available data and understand its patterns more easily.

The project demonstrates the practical application of Java programming, text preprocessing, rule-based sentiment classification, keyword frequency analysis, and web development. The following sections discuss the related work, problem statement, objectives, system workflow, implementation results, software and hardware requirements, conclusion, and future enhancements.

3. LITERATURE SURVEY

3.1 Google Trends

Google Trends is a web-based platform that analyzes the popularity of search terms over time. It allows users to compare topics across different regions and time periods. The platform identifies rising search interests and displays trends through graphs and visualizations. It helps users understand which topics are receiving greater attention. This is related to the trend detection feature of the proposed project.

3.2 GDELT Project

The GDELT Project monitors and analyzes news media from around the world. It provides tools to explore news articles, topics, media coverage, and tone. Its dashboards help users examine how different subjects are reported across news sources. The platform supports data access for further analysis. It is relevant to the proposed project's news analysis and visualization features.

3.3 NewsAPI

NewsAPI provides an interface for retrieving news articles from multiple publishers. Developers can use its API to search for articles based on keywords, sources, and other filters. The retrieved information can be integrated into applications for news collection and display. It can support systems that organize news into a searchable feed. This is related to the news feed component of the proposed project.

3.4 NewsSentimentStreaming

NewsSentimentStreaming is an open-source project that analyzes news sentiment and presents trending sentiment information through a dashboard. It collects news from sources such as NewsAPI and GDELT and processes articles through a streaming pipeline. It includes sentiment scoring, trend aggregation, and visual analytics. Its architecture demonstrates how news

processing can be combined with dashboard visualization. Unlike this project, the proposed application uses a simpler Java backend and a CSV dataset.

3.5 Media Cloud

Media Cloud is a research platform for studying online media and news coverage. It supports the exploration of media sources, topics, and changes in coverage over time. Researchers can use its tools to investigate how news stories are discussed across different publications. Its focus on media analysis makes it relevant to understanding news trends. The proposed project similarly organizes news records and summarizes recurring topics.

3.6 Brandwatch

Brandwatch provides consumer intelligence and social media monitoring tools. Its platform helps organizations study online conversations, identify emerging topics, and understand audience sentiment. Its analytics features support the examination of public discussions and brand-related mentions. These capabilities demonstrate how sentiment and topic analysis can support decision-making. The proposed project applies similar basic concepts to a smaller, predefined dataset.

3.7 Talkwalker

Talkwalker is a social listening and analytics platform that monitors online conversations across digital channels. It helps users analyze topics, mentions, and audience reactions using dashboards and reporting features. Such tools can be used to understand discussion patterns and public opinion. The platform illustrates the value of combining text analysis with visual reporting. The proposed project follows a similar general approach but uses rule-based sentiment classification and keyword frequency analysis.

3.8 Hootsuite

Hootsuite is a social media management platform that helps users manage social accounts and monitor online engagement. Its analytics capabilities provide summaries of audience interactions and social media performance. These summaries help users understand activity and compare engagement patterns. The platform is relevant to the reporting and dashboard aspects of the proposed project. However, the proposed project focuses specifically on sentiment classification and frequently occurring keywords in its dataset.

4. Problem Statement and Objectives

1. Problem Statement

Analyzing large amounts of news and social media content manually is time-consuming and makes it difficult to identify sentiment and trending topics.

2. Problem Description

News and social media data contain valuable information about public opinions and popular topics. Manually analyzing this information makes it difficult to classify sentiment and identify frequently occurring keywords efficiently.

3. Objectives

1. To develop a web-based application for analyzing news and social media content.
2. To load and process textual data from a CSV dataset.
3. To classify content into Positive, Negative, and Neutral sentiments using a rule-based approach.
4. To identify trending topics based on frequently occurring keywords.
5. To display sentiment results, news records, trends, and analytical summaries through a user-friendly dashboard.
6. To simplify data interpretation and help users understand patterns in the available dataset.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Laptop/Desktop Computer
- Minimum 4 GB RAM
- Keyboard and Mouse
- Internet Connection

Software Requirements:

- Java Development Kit (JDK)
- Visual Studio Code
- Java Programming Language
- HTML, CSS, and JavaScript
- CSV Dataset
- Google Chrome Web Browser
- macOS / Windows Operating System

6. IMPLEMENTATION

Module 1: Data Loading (DataLoader.java)

```
public static List<NewsRecord> loadData(String filePath) {
    List<NewsRecord> records = new ArrayList<>();

    try (BufferedReader br = new BufferedReader(
            new FileReader(filePath))) {

        String line;
        br.readLine(); // Skip header

        while ((line = br.readLine()) != null) {
            String[] data = line.split(",", -1);

            if (data.length >= 7) {
                NewsRecord record = new NewsRecord(
                    data[0], data[1], data[2],
                    data[3], data[4], data[5],
                    data[6]
                );
                records.add(record);
```

```
            }
        }
    } catch (IOException e) {
        e.printStackTrace();
    }

    return records;
}
```

Module 2: Text Preprocessing (Preprocessor.java)

```
public static String preprocess(String text) {
    if (text == null) {
        return "";
    }

    text = text.toLowerCase();
    text = text.replaceAll("[^a-z0-9\\s]", " ");
    text = text.replaceAll("\\s+", " ").trim();

    return text;
}
```

Module 3: Sentiment Analysis (SentimentAnalyzer.java)

```
public static String analyzeSentiment(String text) {
    if (text == null || text.trim().isEmpty()) {
        return "Neutral";
    }

    String content = text.toLowerCase();

    String[] positiveWords = {
        "good", "great", "excellent",
        "success", "happy", "positive"
    };

    String[] negativeWords = {
        "bad", "poor", "failure",
        "sad", "negative", "problem"
    };

    int positive = 0;
    int negative = 0;
```

```
    for (String word : positiveWords) {
        if (content.contains(word)) {
            positive++;
        }
    }

    for (String word : negativeWords) {
        if (content.contains(word)) {
            negative++;
        }
    }

    if (positive > negative) {
        return "Positive";
    } else if (negative > positive) {
        return "Negative";
    }

    return "Neutral";
}
```

Module 4: Trending Topic Detection (TrendAnalyzer.java)

```
public static Map<String, Integer> analyzeTrends(
        List<NewsRecord> records) {

    Map<String, Integer> trends = new HashMap<>();

    for (NewsRecord record : records) {
        String text = Preprocessor.preprocess(
            record.getContent()
        );

        for (String word : text.split("\\s+")) {
            if (word.length() > 3) {
                trends.put(
                    word,
                    trends.getOrDefault(word, 0) + 1
                );
            }
        }
    }
    return trends;
}
```

7. RESULTS & DISCUSSION

1. Home Page

The home page provides an overview of the News & Social Media Analytics application and its main features. It allows users to navigate to different sections of the application.

2. News Feed

The News Feed displays news records from the CSV dataset along with their categories and sentiment information. It helps users explore the available news content.

3. Sentiment Analysis

The Sentiment Analysis module classifies news content into Positive, Negative, and Neutral categories. It helps users understand the sentiment distribution in the dataset.

4. Trending Topics

The Trending Topics module identifies frequently occurring keywords in the available dataset. It helps users recognize recurring topics in the collected news content.

5. Analytics Dashboard

The Analytics Dashboard presents summaries of the dataset, including sentiment distribution and category or source information. It makes the analyzed results easier to understand.

8. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The News & Social Media Analytics project successfully demonstrates the analysis of news and social media content using a CSV dataset. The system performs text preprocessing, rule-based sentiment classification, and trending topic detection based on frequently occurring keywords. It presents the results through a user-friendly web dashboard containing a news feed, sentiment analysis, trending topics, and analytics. The project achieves its main objectives by simplifying data analysis and making information easier to understand.

8.2 Future Scope

Short-Term Enhancements:

- Improve the user interface and dashboard design.
- Add more datasets and news categories.
- Provide downloadable analytical reports.

Medium-Term Enhancements:

- Integrate live news APIs for automatic data collection.
- Implement machine-learning models to improve sentiment classification.
- Add date-wise and time-based trend analysis.

Long-Term Enhancements:

- Support multiple languages for sentiment analysis.
- Develop advanced predictive analytics for emerging topics.
- Create real-time monitoring and alert features for trending news.

9. REFERENCES

[1] Google Trends. Available at: https://trends.google.co.in/trends/

[2] GDELT Project. Available at: https://www.gdeltproject.org/

[3] NewsAPI Documentation. Available at: https://newsapi.org/docs

[4] NewsSentimentStreaming Project. Available at: https://github.com/vracks/NewsSentimentStreaming

[5] Media Cloud. Available at: https://mediacloud.org/

[6] Brandwatch. Available at: https://www.brandwatch.com/

[7] Talkwalker. Available at: https://www.talkwalker.com/

[8] Hootsuite. Available at: https://www.hootsuite.com/

[9] Oracle Java Documentation. Available at: https://docs.oracle.com/en/java/

[10] MDN Web Docs — HTML, CSS, and JavaScript. Available at: https://developer.mozilla.org/